Measurement of Laser Beam Waist Using the Knife-Edge Method

Author: Nakarin Jayjong

Knife-edge method

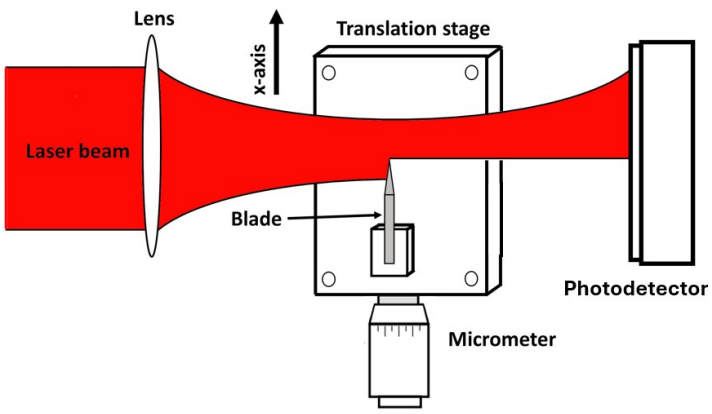

Figure 1. A scheme for the measurement of laser beam waist using the knife-edge technique.

A common laboratory technique for measuring the beam waist of a Gaussian beam is the knife-edge method. The intensity distribution in the x–y plane is

$$I(x,y) = I_0 \exp\left[-\frac{2\big((x-x_0)^2+(y-y_0)^2\big)}{w_0^2}\right], \tag{1}$$

where w_0 is the beam waist, I_0 is the peak intensity, and (x_0, y_0) is the beam center.

In this method, a blade is scanned along the x-axis, partially blocking the beam. The transmitted power is obtained by integrating the intensity over the unblocked region:

$$P(x) = \int_x^{\infty}\int_{-\infty}^{\infty} I(x',y')\,dy'\,dx'. \tag{2}$$

Substituting the Gaussian intensity profile,

$$I(x',y') = I_0 \exp\left[-\frac{2\big((x'-x_0)^2+(y'-y_0)^2\big)}{w_0^2}\right], \tag{3}$$

the integration over y' gives

$$\int_{-\infty}^{\infty} \exp\left[-\frac{2(y'-y_0)^2}{w_0^2}\right] dy' = \frac{w_0}{\sqrt{2}}\sqrt{\pi}. \tag{4}$$

Thus,

$$P(x) = I_0 \frac{w_0}{\sqrt{2}}\sqrt{\pi}\int_x^{\infty} \exp\left[-\frac{2(x'-x_0)^2}{w_0^2}\right] dx'. \tag{5}$$

With the substitution

$$u = \frac{\sqrt{2}(x'-x_0)}{w_0}, \quad dx' = \frac{w_0}{\sqrt{2}}\,du,$$

this becomes

$$P(x) = I_0 \frac{w_0^2}{2}\sqrt{\pi}\int_{\frac{\sqrt{2}(x-x_0)}{w_0}}^{\infty} e^{-u^2}\,du. \tag{6}$$

Using the identity

$$\int_a^{\infty} e^{-u^2}\,du = \frac{\sqrt{\pi}}{2}\left[1-\operatorname{erf}(a)\right], \tag{7}$$

we obtain

$$P(x) = \frac{I_0 \pi w_0^2}{2} \left[1 - \mathrm{erf}\left(\frac{\sqrt{2}(x - x_0)}{w_0} \right) \right]. \tag{8}$$

More generally, allowing for an offset and maximum power scaling, this can be written as [1, 2]

$$P(x) = P_0 + \frac{P_{\max}}{2} \left[1 - \mathrm{erf}\left(\frac{\sqrt{2}(x - x_0)}{w_0} \right) \right]. \tag{9}$$

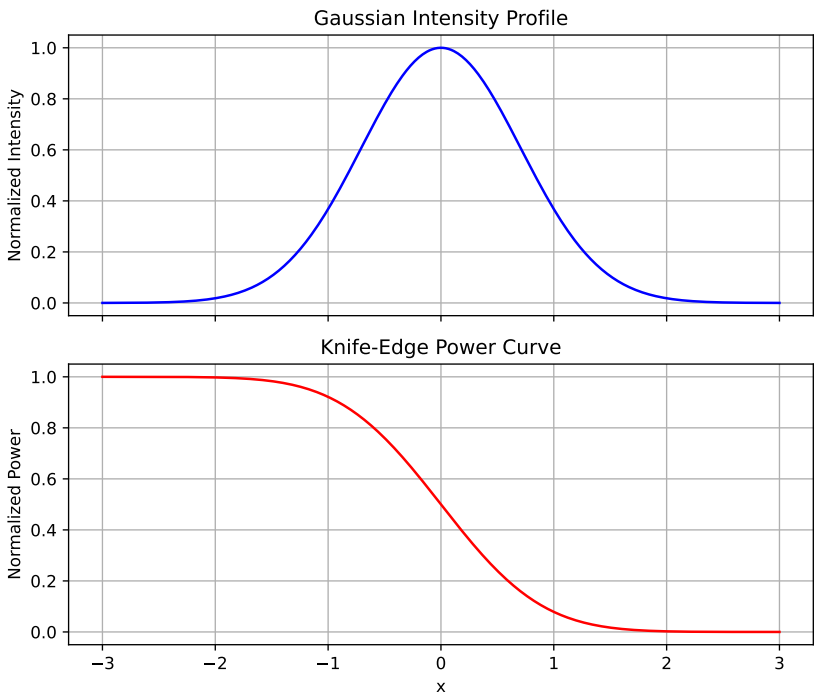

Figure 2. Normalized Gaussian beam intensity and knife-edge power as functions of the x-position. The top panel shows the normalized intensity $I(x)/I_{\max}$, following a Gaussian profile centered at x_0. The bottom panel shows the normalized transmitted power $P(x)$ obtained using the knife-edge method, which follows an error-function dependence. Both curves share the same x-axis for direct comparison of the beam profile and the integrated power.

Python script

Listing 1. Python script to fit knife-edge data and extract Gaussian beam waist

```python
"""
Estimate Gaussian beam radius using knife-edge method
"""
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
from scipy.special import erf
from uncertainties import ufloat

# ----------------------------
# Experimental Data
# ----------------------------

x = np.array([
    0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4,
    4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5
])  # Position (mm)

P = np.array([
    2.05, 2.035, 2.05, 2.047, 2.011, 2.025, 1.955, 1.86, 1.575,
    1.14, 0.705, 0.316, 0.085, 0.016, 0.008, 0.004, 0.002, 0.002
])  # Power (W)

dP = 2*np.array([
    0.005, 0.005, 0.005, 0.005, 0.005, 0.005, 0.010, 0.005, 0.010,
    0.005, 0.005, 0.005, 0.005, 0.001, 0.001, 0.001, 0.001, 0.001
])  # Power uncertainty (W)

# ----------------------------
# Knife-edge Model with Background
# ----------------------------
def knife_edge(x, P0, Pmax, x0, w0):
    """
    P0   : background power (W)
    Pmax : maximum beam power contribution (W)
    x0   : beam center position (mm)
    w0   : 1/e^2 beam waist (mm)
    """
    return P0 + 0.5 * Pmax * (1 - erf(np.sqrt(2) * (x - x0) / w0))

# ----------------------------
# Curve Fitting
# ----------------------------
initial_guess = [0.0, max(P), 4.5, 1.0]  # [P0, Pmax, x0, w0]

popt, pcov = curve_fit(
    knife_edge,
    x,
    P,
    sigma=dP,
    p0=initial_guess,
    absolute_sigma=True
)

P0_fit, Pmax_fit, x0_fit, w0_fit = popt
perr = np.sqrt(np.diag(pcov))

# ----------------------------
# Beam Waist with Uncertainty
# ----------------------------
w0_fit_u = ufloat(w0_fit, perr[3])
resolution = 0.01  # mm, micrometer resolution
w0 = w0_fit_u

# ----------------------------
# Print Results
# ----------------------------
print(f"Background P0 = {P0_fit:.4f}    {perr[0]:.4f} W")
print(f"Maximum beam power Pmax = {Pmax_fit:.4f}    {perr[1]:.4f} W")
print(f"Beam center x0 = {x0_fit:.4f}    {perr[2]:.4f} mm")
print(f"Beam waist w0 = {w0:.1uS} mm")

# ----------------------------
# Plot Data and Fit
# ----------------------------
x_fit = np.linspace(min(x), max(x), 400)
P_fit = knife_edge(x_fit, *popt)

plt.figure(figsize=(7, 5))
plt.errorbar(x, P, yerr=dP, fmt='o', capsize=3, label='Data')
plt.plot(x_fit, P_fit, 'r-', label='Fit')
plt.xlabel('Position (mm)')
```

```
plt.ylabel('Power (W)')
plt.title('Knife-edge Measurement of Beam Waist with Background')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

References

[1] J. Arnaud, W. Hubbard, G. Mandeville, B. De la Claviere, E. Franke, and J. Franke, "Technique for fast measurement of gaussian laser beam parameters," *Applied optics*, vol. 10, no. 12, pp. 2775–2776, 1971.

[2] M. González-Cardel, P. Arguijo, and R. Díaz-Uribe, "Gaussian beam radius measurement with a knife-edge: a polynomial approximation to the inverse error function," *Applied optics*, vol. 52, no. 16, pp. 3849–3855, 2013.